100 Machine Learning & Artificial Intelligence MCQs

Complete MCQ Study Guide with Correct Answers and Explanations

Question → 4 Options → Correct Answer → Explanation → Explanation of all other choices

This edition is fully in English and follows the structure of the provided reference document. It contains 100 questions covering Machine Learning, AI, Generative AI, LLMs, RAG, NLP, Computer Vision, MLOps, evaluation, and AI safety.

How to Use This Guide

Try to answer each question before checking the answer. Then read the explanation to understand the concept rather than memorizing the option number.

Topics Covered

- Machine Learning Fundamentals
- Supervised, Unsupervised and Reinforcement Learning
- Data Splitting, Leakage and Cross-Validation
- Metrics and Confusion Matrix
- Regression, Classification, Trees and Ensembles
- Feature Engineering and Preprocessing
- Neural Networks and Deep Learning
- NLP and Transformers
- Generative AI and LLMs
- Prompt Engineering
- RAG and Embeddings
- AI Agents and Tool Calling
- Computer Vision
- MLOps and Deployment
- AI Security, Privacy, Bias and Responsible AI

Question 1 of 100

What is the definition of overfitting in machine learning?

Option 1: A model has too little training data only.

Option 2: A model always predicts the majority class.

Option 3: A model cannot be trained.

Option 4: A model memorizes training data and generalizes poorly.

✔ Correct Answer: Option 4

■ **Why this is correct:** This is correct because overfitting occurs when a model learns the training examples, including noise, too closely and therefore performs poorly on unseen data.

■ **Why the other options are incorrect:**

Option 1: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 4.

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 4.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 4.

Question 2 of 100

What is the primary purpose of a validation set during the model development lifecycle?

Option 1: Tune and evaluate model choices during development.

Option 2: Replace the training set entirely.

Option 3: Store production logs and user data.

Option 4: Encrypt final model weights for security.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because a validation set is used during development to compare models and tune hyperparameters without using the final test set.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 3 of 100

Why do engineers build specific evaluation datasets for Generative AI applications?

Option 1: To eliminate the need for testing phases.

Option 2: To guarantee that every model answer is factual.

Option 3: To automatically reduce the model parameter size.

Option 4: To measure performance on the behaviors the application actually needs.

✔ Correct Answer: Option 4

■ **Why this is correct:** This is correct because application-specific evaluation datasets measure whether a Generative AI system performs the behaviors and tasks that actually matter in its intended use.

■ **Why the other options are incorrect:**

Option 1: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 4.

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 4.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 4.

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Question 4 of 100

What is the core principle of Transfer Learning?

Option 1: Training a model completely from scratch without data.

Option 2: Adapting a pretrained model to a related task.

Option 3: Removing model parameters randomly to save space.

Option 4: Using two separate production databases simultaneously.

✔ Correct Answer: Option 2

■ **Why this is correct:** This is correct because transfer learning reuses a pretrained model and adapts it to a related task, often reducing training time and data requirements.

■ **Why the other options are incorrect:**

Option 1: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

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Question 5 of 100

Which performance metric is most commonly used for standard classification problems?

Option 1: Accuracy.

Option 2: Mean Squared Error (MSE).

Option 3: R-squared (R^2).

Option 4: BLEU score only.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because accuracy measures the proportion of all classification predictions that are correct and is a common baseline metric for standard balanced classification tasks.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 6 of 100

What causes Data Leakage in a machine learning pipeline?

Option 1: The final model size shrinks automatically.

Option 2: It completely prevents the gradient descent process.

Option 3: Information unavailable at prediction time enters training or evaluation.

Option 4: It only changes the visual user interface (UI) styling.

✔ Correct Answer: Option 3

■ **Why this is correct:** This is correct because data leakage occurs when information that would not be available at real prediction time enters training or evaluation.

■ **Why the other options are incorrect:**

Option 1: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

Question 7 of 100

What is the correct mathematical formula for Precision?

Option 1: TP / (TP + FN)

Option 2: TP / (TP + FP)

Option 3: TN / (TN + FP)

Option 4: (TP + TN) / Total Samples

✔ Correct Answer: Option 2

■ **Why this is correct:** This is correct because precision is the proportion of predicted positives that are actually positive: TP / (TP + FP).

■ **Why the other options are incorrect:**

Option 1: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

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Question 8 of 100

What is a major risk when training a model on a highly imbalanced dataset?

Option 1: A model may favor the majority class.

Option 2: The model will fail to load standard JSON files.

Option 3: The CPU will immediately stop executing Python code.

Option 4: The database will lose its internal storage indexes.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because a highly imbalanced dataset can cause a model to favor the majority class and appear accurate while performing poorly on the minority class.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

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Question 9 of 100

What is the phase called when a fully trained model runs live data to make a prediction?

Option 1: Compilation

Option 2: Serialization

Option 3: Normalization

Option 4: Inference.

✔ Correct Answer: Option 4

■ **Why this is correct:** This is correct because inference is the stage in which a trained model receives new data and produces a prediction.

■ **Why the other options are incorrect:**

Option 1: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 4.

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 4.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 4.

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Question 10 of 100

Which technique randomly deactivates neurons during neural network training to mitigate overfitting?

Option 1: Hashing

Option 2: Indexing

Option 3: Dropout.

Option 4: Sharding

✔ Correct Answer: Option 3

■ **Why this is correct:** This is correct because dropout randomly deactivates units during training and can reduce over-reliance on particular neurons, helping to reduce overfitting.

■ **Why the other options are incorrect:**

Option 1: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

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Question 11 of 100

What defines underfitting in machine learning?

Option 1: A model memorizes training data perfectly.

Option 2: A model is too simple to capture the underlying structure of the data.

Option 3: The training process runs out of system RAM.

Option 4: The model achieves 100% test accuracy.

✔ Correct Answer: Option 2

■ **Why this is correct:** This is correct because underfitting occurs when a model is too simple or constrained to capture the important structure in the data.

■ **Why the other options are incorrect:**

Option 1: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

Question 12 of 100

What is K-Fold Cross-Validation?

Option 1: Passing data through a secure network firewall.

Option 2: Combining two separate neural networks into one architecture.

Option 3: Splitting data into K subsets to train and test the model multiple times on different configurations.

Option 4: Deleting data duplicates automatically.

✔ Correct Answer: Option 3

■ **Why this is correct:** This is correct because K-Fold Cross-Validation divides the data into K folds and repeatedly trains and validates the model using different folds.

■ **Why the other options are incorrect:**

Option 1: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

Question 13 of 100

What is the fundamental difference between Supervised and Unsupervised Learning?

Option 1: Supervised learning doesn't require modern computers.

Option 2: Supervised learning uses labeled training data; unsupervised learning uses unlabeled data.

Option 3: Unsupervised learning is only used for front-end graphic design.

Option 4: Supervised learning never undergoes evaluation.

✔ Correct Answer: Option 2

■ **Why this is correct:** This is correct because supervised learning uses labeled examples, whereas unsupervised learning works without target labels.

■ **Why the other options are incorrect:**

Option 1: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

Question 14 of 100

What is a Hyperparameter in machine learning?

Option 1: Internal weights learned automatically by the model during optimization.

Option 2: The physical transfer speed of the local internet router.

Option 3: Configuration settings explicitly set by the developer before the training process begins.

Option 4: The total number of columns in the raw input CSV file.

✔ Correct Answer: Option 3

■ **Why this is correct:** This is correct because hyperparameters are configuration choices set by the practitioner, such as learning rate, batch size, epochs, or tree depth.

■ **Why the other options are incorrect:**

Option 1: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

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Question 15 of 100

How do Regression tasks differ from Classification tasks?

Option 1: Regression is only executed inside SQL relational databases.

Option 2: Classification deletes old text files from the hard drive automatically.

Option 3: Regression predicts continuous numerical values, while classification predicts categorical labels.

Option 4: They are identical terms for the same exact mathematical formula.

✔ Correct Answer: Option 3

■ **Why this is correct:** This is correct because regression predicts continuous numerical values, while classification predicts discrete categories or labels.

■ **Why the other options are incorrect:**

Option 1: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

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Question 16 of 100

What is Feature Engineering?

Option 1: Designing the graphical layout of a website application.

Option 2: Upgrading physical GPU drivers on a hardware machine.

Option 3: The process of selecting, transforming, and combining raw data fields into meaningful inputs for a model.

Option 4: Automating database backups.

✔ Correct Answer: Option 3

■ **Why this is correct:** This is correct because feature engineering creates useful model inputs by selecting, transforming, encoding, or combining raw data.

■ **Why the other options are incorrect:**

Option 1: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

Question 17 of 100

What do L1 (Lasso) and L2 (Ridge) Regularization techniques do?

Option 1: Compress the final model file into a smaller .zip archive format.

Option 2: Add a penalty term to the loss function to prevent model weights from becoming too large.

Option 3: Encrypt user passwords before database storage.

Option 4: Speed up raw text rendering on data screens.

✔ Correct Answer: Option 2

■ **Why this is correct:** This is correct because L1 and L2 regularization add penalties to the objective function to discourage excessively large model weights and reduce overfitting.

■ **Why the other options are incorrect:**

Option 1: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

Question 18 of 100

What is the core idea behind Ensemble Learning?

Option 1: Writing source code using multiple programming languages simultaneously.

Option 2: Connecting multiple physical servers to a localized hardware network switch.

Option 3: Combining the predictions of multiple individual models to create a single, more robust prediction.

Option 4: Minimizing dataset column sizes.

✔ Correct Answer: Option 3

■ **Why this is correct:** This is correct because ensemble learning combines predictions from multiple models to produce a potentially more robust final prediction.

■ **Why the other options are incorrect:**

Option 1: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

Question 19 of 100

What is the Bias-Variance Tradeoff?

Option 1: Trading CPU computing power for cloud storage space.

Option 2: The balance between a model's systematic error (bias) and its sensitivity to small fluctuations in the training set (variance).

Option 3: The speed difference between uploading data and downloading data.

Option 4: A business negotiation strategy.

✔ Correct Answer: Option 2

■ **Why this is correct:** This is correct because the bias-variance tradeoff concerns balancing systematic model error against sensitivity to variations in the training data.

■ **Why the other options are incorrect:**

Option 1: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

Question 20 of 100

What is a Confusion Matrix?

Option 1: An error message that immediately crashes a running Python script.

Option 2: A security tool used to generate random user passwords.

Option 3: A tabular layout that visualizes the exact counts of True Positives, True Negatives, False Positives, and False Negatives.

Option 4: A corrupted spreadsheet format.

✔ Correct Answer: Option 3

■ **Why this is correct:** This is correct because a confusion matrix summarizes classification results using True Positives, True Negatives, False Positives, and False Negatives.

■ **Why the other options are incorrect:**

Option 1: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 3.

Question 21 of 100

What is Machine Learning?

Option 1: A method where systems learn patterns from data to make predictions or decisions.

Option 2: A process that only stores data in databases.

Option 3: A programming language used only for websites.

Option 4: A hardware component used to increase RAM.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because machine learning enables systems to learn useful patterns from data and use those learned patterns for predictions or decisions.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 22 of 100

What is Artificial Intelligence (AI)?

Option 1: Only a technique for sorting database rows.

Option 2: The field of building systems that perform tasks associated with intelligent behavior.

Option 3: A replacement for every computer operating system.

Option 4: A type of physical computer memory.

✔ Correct Answer: Option 2

■ **Why this is correct:** This is correct because Artificial Intelligence is the broader field concerned with building systems that perform tasks associated with intelligent behavior.

■ **Why the other options are incorrect:**

Option 1: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

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Question 23 of 100

What is Deep Learning?

Option 1: A database backup strategy.

Option 2: A branch of machine learning based heavily on multi-layer neural networks.

Option 3: A method for compressing ZIP files.

Option 4: A programming syntax for SQL.

✔ Correct Answer: Option 2

■ **Why this is correct:** This is correct because deep learning is a subset of machine learning that relies heavily on neural networks with multiple layers.

■ **Why the other options are incorrect:**

Option 1: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

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Question 24 of 100

What is a neural network?

Option 1: A model composed of interconnected computational units that learn parameters from data.

Option 2: A physical network cable.

Option 3: A database table with foreign keys.

Option 4: A password manager.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because neural networks contain interconnected computational units whose parameters are learned from data.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 25 of 100

What is supervised learning?

Option 1: Learning from labeled examples.

Option 2: Learning without any data.

Option 3: Learning only through database indexes.

Option 4: Learning by randomly deleting features.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because supervised learning learns a mapping from inputs to known target labels or values.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 26 of 100

What is unsupervised learning commonly used for?

Option 1: Discovering patterns or groups in unlabeled data.

Option 2: Encrypting model weights.

Option 3: Compiling Python code.

Option 4: Sending emails from a server.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because unsupervised learning is commonly used to discover structure, patterns, or groups in data without target labels.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 27 of 100

What is reinforcement learning?

Option 1: Learning through interaction using rewards and penalties.

Option 2: Learning only from SQL tables.

Option 3: A technique for removing duplicate images.

Option 4: A method for compressing neural networks into ZIP files.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because reinforcement learning learns through interaction with an environment using rewards and penalties or other feedback signals.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

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Question 28 of 100

What is a training set?

Option 1: Data used to learn model parameters.

Option 2: Data permanently hidden from the model for final evaluation only.

Option 3: A database backup.

Option 4: A collection of production passwords.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because the training set is the data used to learn the model's parameters.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

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Question 29 of 100

What is a test set?

Option 1: Data used for final unbiased evaluation after model development.

Option 2: Data used to update weights every training step.

Option 3: A list of hyperparameters.

Option 4: A database index.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because the test set is reserved for final evaluation after model development and tuning are complete.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 30 of 100

What is a model parameter?

Option 1: A value learned from data during model training.

Option 2: A setting chosen only after deployment.

Option 3: A database table name.

Option 4: A UI color.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because model parameters, such as weights and biases, are values learned from data during optimization.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 31 of 100

Which is an example of a hyperparameter?

Option 1: A learned neural-network weight.

Option 2: Learning rate.

Option 3: A prediction generated by the model.

Option 4: A row in the test dataset.

✔ Correct Answer: Option 2

■ **Why this is correct:** This is correct because learning rate is a hyperparameter that controls the size of parameter updates during optimization.

■ **Why the other options are incorrect:**

Option 1: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 2.

Question 32 of 100

What does an epoch mean in model training?

Option 1: One complete pass through the training dataset.

Option 2: One database transaction.

Option 3: One feature column.

Option 4: One prediction only.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because one epoch represents one complete pass through the training dataset.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 33 of 100

What does batch size control?

Option 1: The number of training examples processed before a parameter update.

Option 2: The number of classes in the dataset.

Option 3: The number of test files stored on disk.

Option 4: The model's final file extension.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because batch size determines how many training examples are processed before a parameter update.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 34 of 100

What is a loss function?

Option 1: A function that quantifies how wrong a model's predictions are.

Option 2: A tool for compressing datasets.

Option 3: A database backup scheduler.

Option 4: A network firewall rule.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because a loss function quantifies the error between predictions and target values and provides an optimization objective.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 35 of 100

What is gradient descent used for?

Option 1: Updating model parameters to minimize a loss function.

Option 2: Creating database indexes.

Option 3: Encrypting network traffic.

Option 4: Generating random passwords.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because gradient descent updates model parameters in a direction intended to reduce the loss function.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 36 of 100

What happens if the learning rate is too high?

Option 1: Training may overshoot good solutions or become unstable.

Option 2: The model automatically becomes perfect.

Option 3: The dataset becomes labeled.

Option 4: The CPU gains more cores.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because an excessively large learning rate can cause optimization to overshoot good solutions or become unstable.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 37 of 100

What happens if the learning rate is too low?

Option 1: Training may become very slow.

Option 2: The model always reaches 100% accuracy.

Option 3: All labels are removed.

Option 4: The dataset becomes encrypted.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because an excessively small learning rate can make optimization very slow and require many updates.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 38 of 100

What is recall in binary classification?

Option 1: TP / (TP + FN)

Option 2: TP / (TP + FP)

Option 3: TN / (TN + FP)

Option 4: (TP + TN) / Total Samples

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because recall is the proportion of actual positive examples that are correctly identified: TP / (TP + FN).

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

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Question 39 of 100

What is F1-score?

Option 1: The harmonic mean of precision and recall.

Option 2: The average of all feature values.

Option 3: The number of training epochs.

Option 4: The model file size in megabytes.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because F1-score is the harmonic mean of precision and recall and summarizes their balance.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

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Question 40 of 100

When is accuracy potentially misleading?

Option 1: When classes are highly imbalanced.

Option 2: When the dataset has column names.

Option 3: When Python is installed.

Option 4: When a model has a loss function.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because accuracy can be misleading on highly imbalanced datasets when a majority-class prediction achieves a high score without identifying the minority class.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 41 of 100

What is specificity?

Option 1: TN / (TN + FP)

Option 2: TP / (TP + FN)

Option 3: TP / (TP + FP)

Option 4: TP / Total Samples

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because specificity measures the proportion of actual negative examples correctly identified: TN / (TN + FP).

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 42 of 100

What is a false positive?

Option 1: The model predicts positive when the actual class is negative.

Option 2: The model predicts negative when the actual class is positive.

Option 3: The model predicts positive and it is positive.

Option 4: The model predicts negative and it is negative.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because a false positive occurs when the model predicts the positive class for an example whose actual class is negative.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 43 of 100

What is a false negative?

Option 1: The model predicts negative when the actual class is positive.

Option 2: The model predicts positive when the actual class is negative.

Option 3: The model predicts positive and it is positive.

Option 4: The model predicts negative and it is negative.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because a false negative occurs when the model predicts the negative class for an example whose actual class is positive.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 44 of 100

What is normalization commonly used for?

Option 1: Putting numeric features onto a comparable scale.

Option 2: Creating new labels automatically.

Option 3: Deleting the validation set.

Option 4: Encrypting model files.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because normalization or scaling places numeric features on more comparable ranges, which can help many machine-learning algorithms.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 45 of 100

What is standardization commonly based on?

Option 1: Subtracting the mean and dividing by the standard deviation.

Option 2: Deleting all negative values.

Option 3: Converting every value to a string.

Option 4: Sorting rows alphabetically.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because standardization commonly transforms a value using (x - mean) / standard deviation.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 46 of 100

Why should preprocessing parameters usually be fitted on the training data only?

Option 1: To avoid leaking information from validation/test data.

Option 2: To make the model larger.

Option 3: To remove all classes.

Option 4: To guarantee zero loss.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because preprocessing parameters should be fitted using training data only to prevent information from validation or test data leaking into the pipeline.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

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Question 47 of 100

What is one-hot encoding used for?

Option 1: Representing categorical values as binary indicator columns.

Option 2: Encrypting passwords.

Option 3: Compressing images.

Option 4: Calculating gradients.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because one-hot encoding represents categorical values using binary indicator columns.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

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Question 48 of 100

What is a categorical feature?

Option 1: A feature representing discrete groups or labels such as color or country.

Option 2: A feature that must always be a floating-point number.

Option 3: A database password.

Option 4: A GPU temperature reading only.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because a categorical feature represents discrete groups or labels such as color, country, or product type.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

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Question 49 of 100

What is dimensionality reduction?

Option 1: Reducing the number of input features while trying to preserve useful information.

Option 2: Increasing every feature value by 10.

Option 3: Adding more database tables.

Option 4: Increasing model output classes.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because dimensionality reduction decreases the number of input dimensions while attempting to preserve useful information.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

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Question 50 of 100

What is PCA?

Option 1: Principal Component Analysis, a dimensionality-reduction technique.

Option 2: A password cracking algorithm.

Option 3: A web framework.

Option 4: A database backup format.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because PCA, or Principal Component Analysis, is a well-known dimensionality-reduction technique.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

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Question 51 of 100

What is clustering?

Option 1: Grouping similar data points without requiring predefined labels.

Option 2: Encrypting data into clusters of files.

Option 3: Training only on labeled examples.

Option 4: Deleting duplicate database rows.

✔ **Correct Answer: Option 1**

■ **Why this is correct:** This is correct because clustering groups similar data points without requiring predefined target labels.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 52 of 100

Which algorithm is a well-known clustering method?

Option 1: K-Means

Option 2: Linear Regression

Option 3: Naive Bayes

Option 4: Logistic Regression

✔ **Correct Answer: Option 1**

■ **Why this is correct:** This is correct because K-Means is a widely used clustering algorithm that partitions data into K groups based on similarity or distance.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 53 of 100

What is linear regression used for?

Option 1: Predicting a continuous numeric target.

Option 2: Generating database passwords.

Option 3: Grouping unlabeled images only.

Option 4: Encrypting text.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because linear regression is commonly used to predict a continuous numerical target.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 54 of 100

What is logistic regression commonly used for?

Option 1: Classification, especially binary classification.

Option 2: Image file compression.

Option 3: Database indexing.

Option 4: GPU overclocking.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because logistic regression is commonly used for classification, particularly binary classification.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 55 of 100

What is a decision tree?

Option 1: A model that makes predictions through a sequence of feature-based decisions.

Option 2: A physical tree-shaped network cable.

Option 3: A database schema diagram only.

Option 4: A neural-network activation function.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because a decision tree makes predictions by applying a sequence of feature-based decision rules.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

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Question 56 of 100

What is Random Forest?

Option 1: An ensemble of decision trees whose predictions are combined.

Option 2: A single linear regression equation.

Option 3: A database replication protocol.

Option 4: A text tokenizer.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because Random Forest is an ensemble of decision trees whose predictions are combined.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

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Question 57 of 100

What is boosting?

Option 1: Building an ensemble sequentially so later models focus on errors made by earlier models.

Option 2: Increasing computer fan speed.

Option 3: Adding database indexes.

Option 4: Increasing image resolution only.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because boosting builds an ensemble sequentially, with later learners focusing on errors or residuals from earlier learners.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

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Question 58 of 100

What is bagging?

Option 1: Training multiple models on bootstrap samples and aggregating their predictions.

Option 2: Encrypting model checkpoints.

Option 3: Deleting all training samples.

Option 4: Converting text to speech.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because bagging trains multiple models on bootstrap samples and aggregates their predictions, often reducing variance.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

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Question 59 of 100

What is regularization used for?

Option 1: Reducing overfitting by constraining model complexity.

Option 2: Guaranteeing perfect test accuracy.

Option 3: Increasing the number of labels.

Option 4: Replacing the dataset.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because regularization constrains model complexity or penalizes complex solutions to reduce overfitting.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

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Question 60 of 100

What is early stopping?

Option 1: Stopping training when validation performance stops improving.

Option 2: Deleting the training dataset early.

Option 3: Stopping the computer before it boots.

Option 4: Removing all model layers.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because early stopping can stop training when validation performance stops improving, helping prevent unnecessary overfitting.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 61 of 100

What is data augmentation?

Option 1: Creating varied training examples from existing data using valid transformations.

Option 2: Deleting half of the dataset.

Option 3: Encrypting labels.

Option 4: Replacing the model with a database.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because data augmentation creates varied training examples from existing data using valid transformations.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 62 of 100

Why is train/validation/test splitting important?

Option 1: It separates learning, model selection, and final evaluation roles.

Option 2: It guarantees the dataset has no noise.

Option 3: It removes the need for metrics.

Option 4: It makes every model deterministic.

✔ **Correct Answer: Option 1**

■ **Why this is correct:** This is correct because separate training, validation, and test sets provide distinct roles for learning, model selection, and final evaluation.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

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Question 63 of 100

What is cross-validation especially useful for?

Option 1: Estimating model performance more robustly when data is limited.

Option 2: Encrypting datasets.

Option 3: Increasing GPU memory.

Option 4: Replacing all preprocessing.

✔ **Correct Answer: Option 1**

■ **Why this is correct:** This is correct because cross-validation provides a more robust performance estimate by evaluating a model across multiple data splits.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

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Question 64 of 100

What is a baseline model?

Option 1: A simple reference approach used to compare more advanced models against.

Option 2: The final production model by definition.

Option 3: A database backup.

Option 4: A hardware cooling system.

✔ **Correct Answer: Option 1**

■ **Why this is correct:** This is correct because a baseline model provides a simple reference point against which more advanced approaches can be compared.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

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Question 65 of 100

What is model generalization?

Option 1: The ability to perform well on unseen data.

Option 2: The ability to memorize training samples.

Option 3: The ability to save files as JSON.

Option 4: The ability to increase CPU speed.

✔ **Correct Answer: Option 1**

■ **Why this is correct:** This is correct because generalization is the ability of a model to perform well on data it did not see during training.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

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Question 66 of 100

What is a feature in machine learning?

Option 1: An input variable used by a model to make predictions.

Option 2: Always the final prediction.

Option 3: A database password.

Option 4: A model's test score only.

✔ **Correct Answer: Option 1**

■ **Why this is correct:** This is correct because a feature is an input variable or attribute used by a model to make predictions.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 67 of 100

What is a label or target?

Option 1: The desired output the model is trained to predict.

Option 2: A GPU driver.

Option 3: A database index.

Option 4: A random feature name.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because a label or target is the desired output that a supervised-learning model is trained to predict.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 68 of 100

What is an outlier?

Option 1: A data point that is unusually far from the general pattern.

Option 2: The most common class always.

Option 3: A model parameter.

Option 4: A database schema.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because an outlier is a data point that is unusually far from the general pattern of the data.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 69 of 100

What is data preprocessing?

Option 1: Preparing raw data for modeling through cleaning, transformation, and related steps.

Option 2: Deploying a model to production only.

Option 3: Writing marketing content.

Option 4: Creating a GPU physically.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because data preprocessing prepares raw data for modeling through cleaning, transformation, encoding, scaling, and related operations.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 70 of 100

What is Generative AI?

Option 1: AI that can generate new content such as text, images, audio, or code.

Option 2: AI that only sorts spreadsheets.

Option 3: A database backup system.

Option 4: A physical GPU architecture.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because Generative AI refers to systems that can generate new content such as text, images, audio, video, or code.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 71 of 100

What is an LLM?

Option 1: Large Language Model.

Option 2: Local Learning Machine.

Option 3: Logical Label Manager.

Option 4: Large Linear Memory.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because LLM stands for Large Language Model.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

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Question 72 of 100

What is tokenization in NLP?

Option 1: Splitting text into tokens that a language model can process.

Option 2: Encrypting text with a password.

Option 3: Converting all text into images.

Option 4: Deleting punctuation from a database.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because tokenization converts text into tokens that a language model can process.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

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Question 73 of 100

What is a prompt?

Option 1: An instruction or input given to a generative AI model.

Option 2: A database primary key.

Option 3: A GPU driver.

Option 4: A model's learned weight.

✔ **Correct Answer: Option 1**

■ **Why this is correct:** This is correct because a prompt is an instruction or input supplied to a generative AI model.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

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Question 74 of 100

What is prompt engineering?

Option 1: Designing and refining prompts to obtain more useful model behavior.

Option 2: Programming GPU firmware.

Option 3: Creating database indexes.

Option 4: Compressing training images.

✔ **Correct Answer: Option 1**

■ **Why this is correct:** This is correct because prompt engineering involves designing and refining prompts to obtain more useful and reliable model behavior.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

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Question 75 of 100

What is a hallucination in generative AI?

Option 1: A generated output that contains fabricated or unsupported information presented as if it were true.

Option 2: A model that has no parameters.

Option 3: A database query that runs quickly.

Option 4: A perfectly verified answer.

✔ **Correct Answer: Option 1**

■ **Why this is correct:** This is correct because an AI hallucination is fabricated, unsupported, or incorrect information presented by a generative model as though it were reliable.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 76 of 100

What is RAG?

Option 1: Retrieval-Augmented Generation, which retrieves relevant information before generation.

Option 2: Random Algorithm Generator.

Option 3: A GPU scheduling protocol.

Option 4: A database compression format.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because RAG, or Retrieval-Augmented Generation, retrieves relevant information and supplies it to a generative model as context.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 77 of 100

What is an embedding in AI/RAG systems?

Option 1: A numerical vector representation capturing useful semantic relationships.

Option 2: A database backup file.

Option 3: A model password.

Option 4: An HTML tag.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because embeddings represent data such as text as numerical vectors that capture useful semantic relationships.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 78 of 100

What is a vector database commonly used for?

Option 1: Storing and searching vector embeddings efficiently.

Option 2: Compiling Python programs.

Option 3: Managing CPU temperature.

Option 4: Encrypting all emails.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because vector databases are designed to store and efficiently search vector embeddings, often using similarity search.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 79 of 100

What is semantic search?

Option 1: Searching based on meaning or intent rather than exact keyword matching only.

Option 2: Searching only filenames alphabetically.

Option 3: Searching a database by primary key only.

Option 4: Searching for duplicate passwords.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because semantic search retrieves information based on meaning or intent rather than relying only on exact keyword matching.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 80 of 100

What is chunking in a RAG pipeline?

Option 1: Splitting documents into smaller pieces for retrieval.

Option 2: Deleting documents permanently.

Option 3: Encrypting the vector database.

Option 4: Changing a model's weights.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because chunking divides long documents into smaller pieces that can be embedded and retrieved efficiently.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 81 of 100

What is context in an LLM application?

Option 1: Information provided to the model that helps it produce a relevant response.

Option 2: Only the model's GPU temperature.

Option 3: A database password.

Option 4: A file extension.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because context is information supplied to a model that helps it produce a relevant response.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 82 of 100

What is a context window?

Option 1: The amount of input/output token context a model can handle in a request.

Option 2: A graphical window used to resize images.

Option 3: A database transaction window.

Option 4: A CPU fan setting.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because a context window is the amount of token context a model can process within a request.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 83 of 100

What is fine-tuning?

Option 1: Further training a pretrained model on task- or domain-specific data.

Option 2: Changing a computer's monitor brightness.

Option 3: Deleting model weights.

Option 4: Creating a database index.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because fine-tuning continues training a pretrained model on task-specific or domain-specific data.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 84 of 100

How does RAG differ from fine-tuning?

Option 1: RAG supplies retrieved information at inference time; fine-tuning changes learned parameters through additional training.

Option 2: They are exactly the same process.

Option 3: Fine-tuning only changes UI colors.

Option 4: RAG requires deleting the base model.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because RAG supplies retrieved information at inference time, while fine-tuning changes learned parameters through additional training.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

■■

Question 85 of 100

What is an AI agent?

Option 1: A system that can perceive context, reason, and take actions toward a goal using available tools or workflows.

Option 2: A static image file.

Option 3: A database table.

Option 4: A GPU memory chip.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because an AI agent can use context, reasoning, tools, and actions to pursue a goal through one or more steps.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

■■

Question 86 of 100

What is tool calling in an AI agent?

Option 1: Allowing a model to request execution of defined external functions or tools.

Option 2: Calling a human employee by phone only.

Option 3: Changing the model's source code automatically.

Option 4: Deleting the model's context.

✔ **Correct Answer: Option 1**

■ **Why this is correct:** This is correct because tool calling allows a model to request execution of explicitly defined external functions or tools.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

■■

Question 87 of 100

What is multimodal AI?

Option 1: AI that can work with multiple modalities such as text, images, audio, or video.

Option 2: AI that uses only one numeric feature.

Option 3: A database with multiple tables.

Option 4: A CPU with multiple cores only.

✔ **Correct Answer: Option 1**

■ **Why this is correct:** This is correct because multimodal AI works with multiple modalities such as text, images, audio, or video.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

■■

Question 88 of 100

What is NLP?

Option 1: Natural Language Processing.

Option 2: Neural Logic Protocol.

Option 3: Network Learning Processor.

Option 4: Natural Label Prediction.

✔ **Correct Answer: Option 1**

■ **Why this is correct:** This is correct because NLP stands for Natural Language Processing, the field concerned with processing and understanding human language.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

■■

Question 89 of 100

What is computer vision?

Option 1: The field of AI concerned with understanding images and video.

Option 2: A method for database indexing.

Option 3: A technique for encrypting passwords.

Option 4: A programming language.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because computer vision focuses on extracting useful information and understanding images and video.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

■■

Question 90 of 100

What is CNN commonly used for?

Option 1: Processing spatial patterns in images and other grid-like data.

Option 2: Managing SQL transactions only.

Option 3: Encrypting text.

Option 4: Scheduling operating-system processes.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because CNNs are neural-network architectures that are particularly effective at learning spatial patterns in images and other grid-like data.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

■■

Question 91 of 100

What is an RNN designed to handle well?

Option 1: Sequential or temporal data where order can matter.

Option 2: Only static database tables.

Option 3: GPU hardware repair.

Option 4: Password hashing only.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because RNNs are designed to process sequential or temporal data where order can matter.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

■■

Question 92 of 100

What is a Transformer architecture known for?

Option 1: Using attention mechanisms to model relationships among tokens.

Option 2: Only performing database backups.

Option 3: Replacing all neural networks with decision trees.

Option 4: Managing computer power supplies.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because Transformer architectures use attention mechanisms to model relationships among tokens and form the basis of many modern language models.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 93 of 100

What is self-attention?

Option 1: A mechanism that lets each token weigh the relevance of other tokens in the same sequence.

Option 2: A technique for encrypting a password with itself.

Option 3: A database join.

Option 4: A method for increasing RAM.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because self-attention allows each token to weigh the relevance of other tokens in the same sequence.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 94 of 100

What is an attention mechanism useful for?

Option 1: Focusing computation on relevant parts of an input.

Option 2: Deleting irrelevant database rows automatically.

Option 3: Increasing monitor brightness.

Option 4: Changing Python syntax.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because attention mechanisms allow models to focus computationally on the most relevant parts of an input.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 95 of 100

What is an activation function such as ReLU used for?

Option 1: Introducing non-linearity into a neural network.

Option 2: Storing database records.

Option 3: Encrypting model files.

Option 4: Generating CSV headers.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because activation functions such as ReLU introduce non-linearity into neural networks.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 96 of 100

What is softmax commonly used for?

Option 1: Converting logits into a probability distribution across classes.

Option 2: Compressing model files.

Option 3: Creating database indexes.

Option 4: Removing all model parameters.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because softmax converts logits into a probability distribution across classes and is commonly used for multiclass classification.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 97 of 100

What is an optimizer in deep learning?

Option 1: An algorithm that updates model parameters based on gradients and the objective.

Option 2: A database storage engine.

Option 3: A text file extension.

Option 4: A computer monitor.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because an optimizer updates model parameters using gradients and an objective such as a loss function.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 98 of 100

What is Adam?

Option 1: A widely used gradient-based optimization algorithm.

Option 2: A database language.

Option 3: An image file format.

Option 4: A network protocol.

✔ Correct Answer: Option 1

■ **Why this is correct:** This is correct because Adam is a widely used gradient-based optimization algorithm with adaptive parameter updates.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Question 99 of 100

What is a learning curve?

Option 1: A plot showing model performance or loss as training progresses or data increases.

Option 2: A graph of CPU fan speed only.

Option 3: A database schema diagram.

Option 4: A network cable specification.

✔ **Correct Answer: Option 1**

■ **Why this is correct:** This is correct because a learning curve shows model performance or loss as training progresses or as the amount of training data changes.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

■■■

Question 100 of 100

What is model drift?

Option 1: A change in real-world data or relationships that can degrade model performance over time.

Option 2: A model file moving to another folder.

Option 3: A database table being renamed.

Option 4: A GPU physically changing position.

✔ **Correct Answer: Option 1**

■ **Why this is correct:** This is correct because a final untouched test set provides a more honest estimate of generalization after model development and tuning are complete.

■ **Why the other options are incorrect:**

Option 2: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 3: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

Option 4: This option is incorrect because it does not describe the concept asked in the question. The correct choice is Option 1.

■■■

Final Answer Key — 1 to 100

1 → 4	2 → 1	3 → 4	4 → 2	5 → 1	6 → 3	7 → 2	8 → 1	9 → 4	10 → 3
11 → 2	12 → 3	13 → 2	14 → 3	15 → 3	16 → 3	17 → 2	18 → 3	19 → 2	20 → 3
21 → 1	22 → 2	23 → 2	24 → 1	25 → 1	26 → 1	27 → 1	28 → 1	29 → 1	30 → 1
31 → 2	32 → 1	33 → 1	34 → 1	35 → 1	36 → 1	37 → 1	38 → 1	39 → 1	40 → 1
41 → 1	42 → 1	43 → 1	44 → 1	45 → 1	46 → 1	47 → 1	48 → 1	49 → 1	50 → 1
51 → 1	52 → 1	53 → 1	54 → 1	55 → 1	56 → 1	57 → 1	58 → 1	59 → 1	60 → 1
61 → 1	62 → 1	63 → 1	64 → 1	65 → 1	66 → 1	67 → 1	68 → 1	69 → 1	70 → 1
71 → 1	72 → 1	73 → 1	74 → 1	75 → 1	76 → 1	77 → 1	78 → 1	79 → 1	80 → 1
81 → 1	82 → 1	83 → 1	84 → 1	85 → 1	86 → 1	87 → 1	88 → 1	89 → 1	90 → 1
91 → 1	92 → 1	93 → 1	94 → 1	95 → 1	96 → 1	97 → 1	98 → 1	99 → 1	100 → 1